HR PORTFOLIO ANALYTICS

IBM HR DATASET

AHMAD AMIR HAMZAH

2026

Hello,

I'm Ahmad Amir Hamzah

Bachelor of Economics

An economics graduate interested in developing myself in data analysis. Over the past few months, I've been independently learning data processing using Excel.

Raw Data - IBM HR Analytics Employee Attrition & Performance

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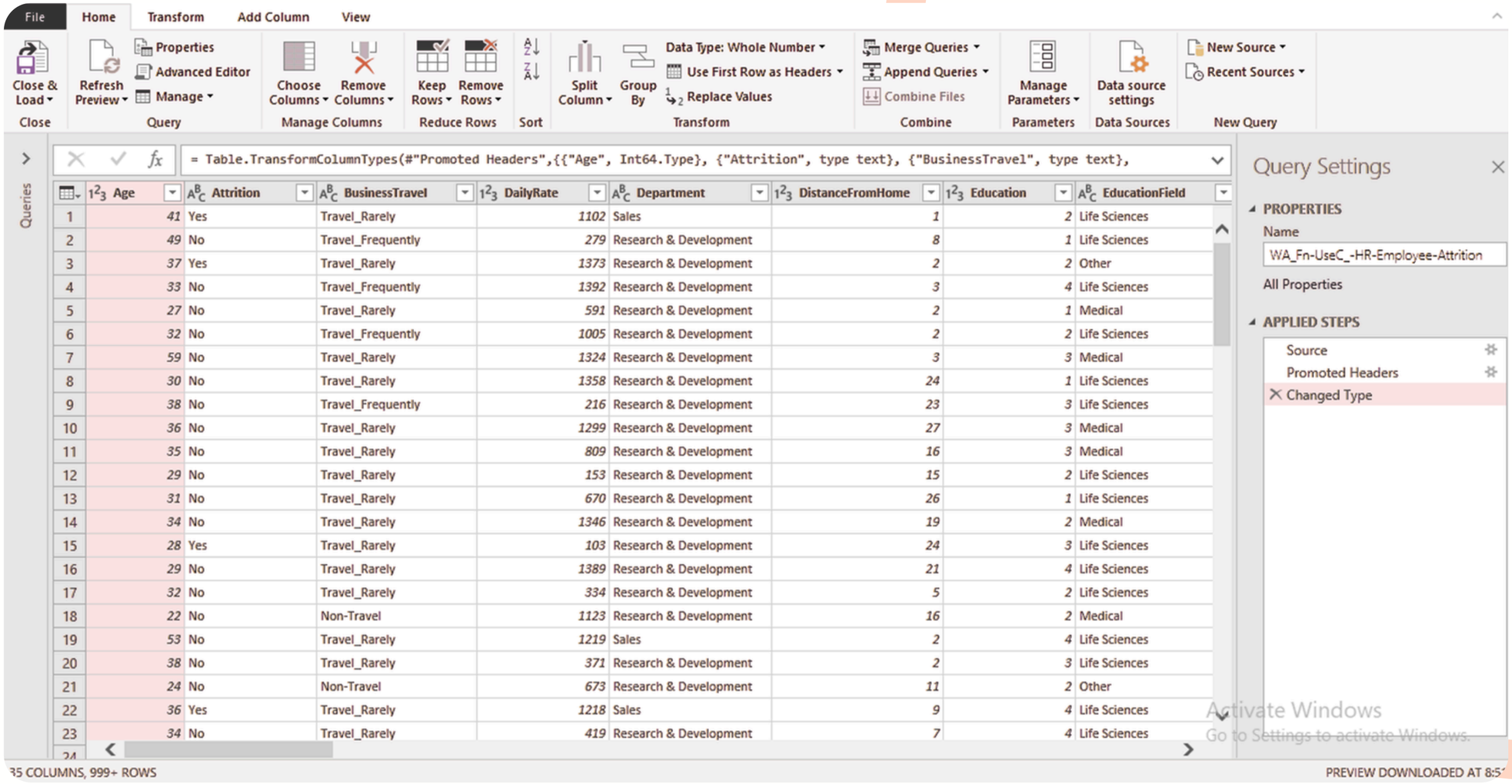


	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField
1	41	Yes	Travel_Rarely	1102	Sales	1	2	Life Sciences
2	49	No	Travel_Frequently	279	Research & Development	8	1	Life Sciences
3	37	Yes	Travel_Rarely	1373	Research & Development	2	2	Other
4	33	No	Travel_Frequently	1392	Research & Development	3	4	Life Sciences
5	27	No	Travel_Rarely	591	Research & Development	2	1	Medical
6	32	No	Travel_Frequently	1005	Research & Development	2	2	Life Sciences
7	59	No	Travel_Rarely	1324	Research & Development	3	3	Medical
8	30	No	Travel_Rarely	1358	Research & Development	24	1	Life Sciences
9	38	No	Travel_Frequently	216	Research & Development	23	3	Life Sciences
10	36	No	Travel_Rarely	1299	Research & Development	27	3	Medical
11	35	No	Travel_Rarely	809	Research & Development	16	3	Medical
12	29	No	Travel_Rarely	153	Research & Development	15	2	Life Sciences
13	31	No	Travel_Rarely	670	Research & Development	26	1	Life Sciences
14	34	No	Travel_Rarely	1346	Research & Development	19	2	Medical
15	28	Yes	Travel_Rarely	103	Research & Development	24	3	Life Sciences
16	29	No	Travel_Rarely	1389	Research & Development	21	4	Life Sciences
17	32	No	Travel_Rarely	334	Research & Development	5	2	Life Sciences
18	22	No	Non-Travel	1123	Research & Development	16	2	Medical
19	53	No	Travel_Rarely	1219	Sales	2	4	Life Sciences
20	38	No	Travel_Rarely	371	Research & Development	2	3	Life Sciences
21	24	No	Non-Travel	673	Research & Development	11	2	Other
22	36	Yes	Travel_Rarely	1218	Sales	9	4	Life Sciences
23	34	No	Travel_Rarely	419	Research & Development	7	4	Life Sciences

Preview Data - IBM HR Analytics Employee Attrition & Performance

Age	Attrition	AttritionFl	AttritionFl	BusinessTravel	DailyRate	Department	DistanceFr	Education	EducationField
41	Yes	1	0	Travel_Rarely	1102	Sales	1	College	Life Sciences
49	No	0	1	Travel_Frequently	279	Research & Development	8	Below College	Life Sciences
37	Yes	1	0	Travel_Rarely	1373	Research & Development	2	College	Other
33	No	0	1	Travel_Frequently	1392	Research & Development	3	Master	Life Sciences
27	No	0	1	Travel_Rarely	591	Research & Development	2	Below College	Medical
32	No	0	1	Travel_Frequently	1005	Research & Development	2	College	Life Sciences
59	No	0	1	Travel_Rarely	1324	Research & Development	3	Bachelor	Medical
30	No	0	1	Travel_Rarely	1358	Research & Development	24	Below College	Life Sciences
38	No	0	1	Travel_Frequently	216	Research & Development	23	Bachelor	Life Sciences
36	No	0	1	Travel_Rarely	1299	Research & Development	27	Bachelor	Medical
35	No	0	1	Travel_Rarely	809	Research & Development	16	Bachelor	Medical
29	No	0	1	Travel_Rarely	153	Research & Development	15	College	Life Sciences
31	No	0	1	Travel_Rarely	670	Research & Development	26	Below College	Life Sciences
34	No	0	1	Travel_Rarely	1346	Research & Development	19	College	Medical
28	Yes	1	0	Travel_Rarely	103	Research & Development	24	Bachelor	Life Sciences
29	No	0	1	Travel_Rarely	1389	Research & Development	21	Master	Life Sciences
32	No	0	1	Travel_Rarely	334	Research & Development	5	College	Life Sciences
22	No	0	1	Non-Travel	1123	Research & Development	16	College	Medical
53	No	0	1	Travel_Rarely	1219	Sales	2	Master	Life Sciences
38	No	0	1	Travel_Rarely	371	Research & Development	2	Bachelor	Life Sciences
24	No	0	1	Non-Travel	673	Research & Development	11	College	Other
36	Yes	1	0	Travel_Rarely	1218	Sales	9	Master	Life Sciences
34	No	0	1	Travel_Rarely	419	Research & Development	7	Master	Life Sciences
21	No	0	1	Travel_Rarely	391	Research & Development	15	College	Life Sciences
34	Yes	1	0	Travel_Rarely	699	Research & Development	6	Below College	Medical
53	No	0	1	Travel_Rarely	1282	Research & Development	5	Bachelor	Other
32	Yes	1	0	Travel_Frequently	1125	Research & Development	16	Below College	Life Sciences
42	No	0	1	Travel_Rarely	691	Sales	8	Master	Marketing
44	No	0	1	Travel_Rarely	477	Research & Development	7	Master	Medical
46	No	0	1	Travel_Rarely	705	Sales	2	Master	Marketing
33	No	0	1	Travel_Rarely	924	Research & Development	2	Bachelor	Medical
44	No	0	1	Travel_Rarely	1459	Research & Development	10	Master	Other

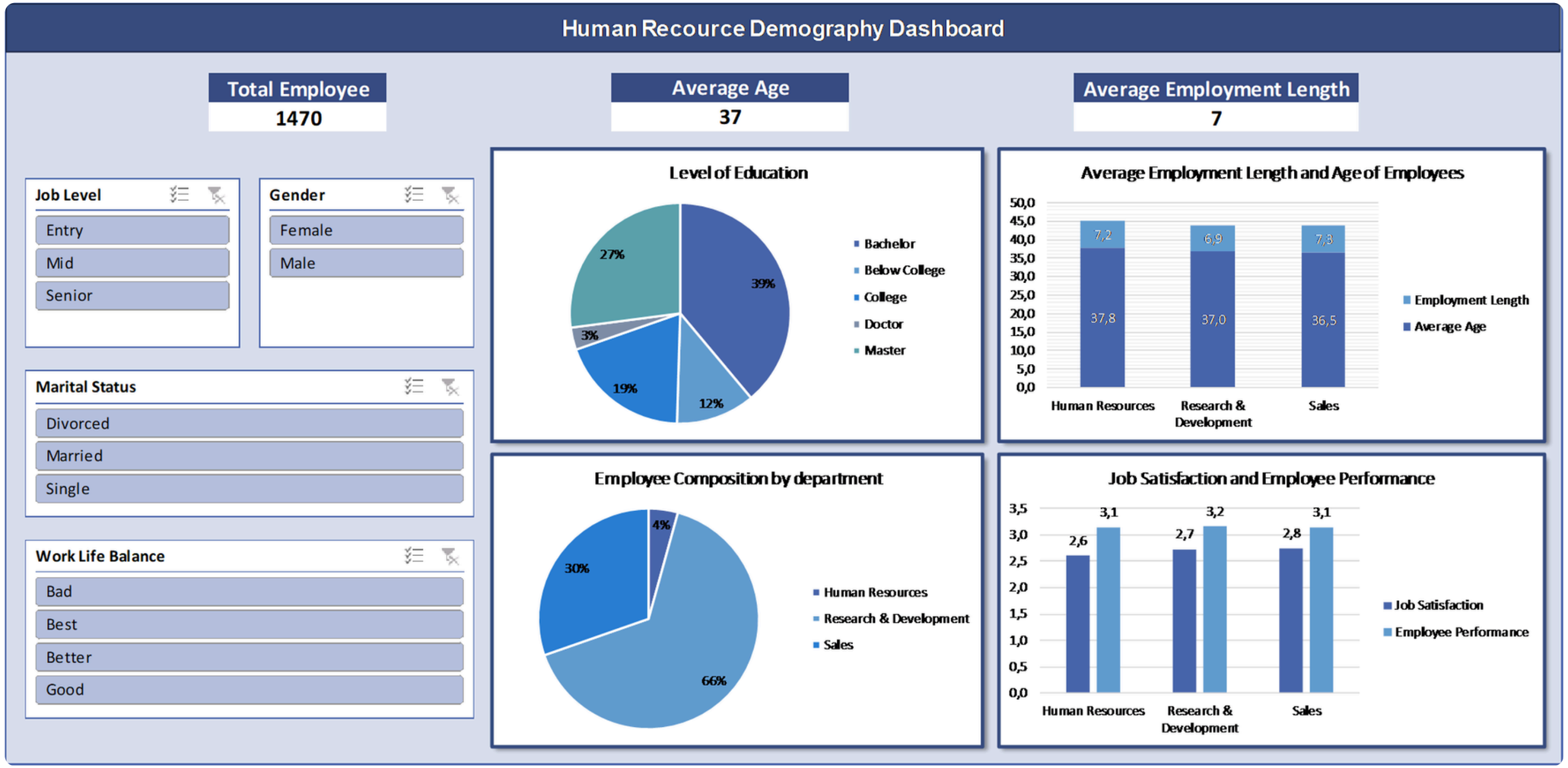
Deskripsi:

Dataset ini berisi data karyawan dari sebuah perusahaan dengan 1.470 baris dan 35 kolom. Data mencakup informasi demografi, pekerjaan, serta faktor-faktor yang berkaitan dengan kepuasan kerja dan status resign.

Link Dataset:

IBM HR Analytics Employee Attrition & Performance

Dashboard 1 - Human Resource Demography



Dashboard 1 Insight

Key Performance Indicator 📊

- Total Karyawan: 1.470
- Rata-rata Umur: 36,9 tahun
- Rata-rata Masa Kerja: 7 tahun

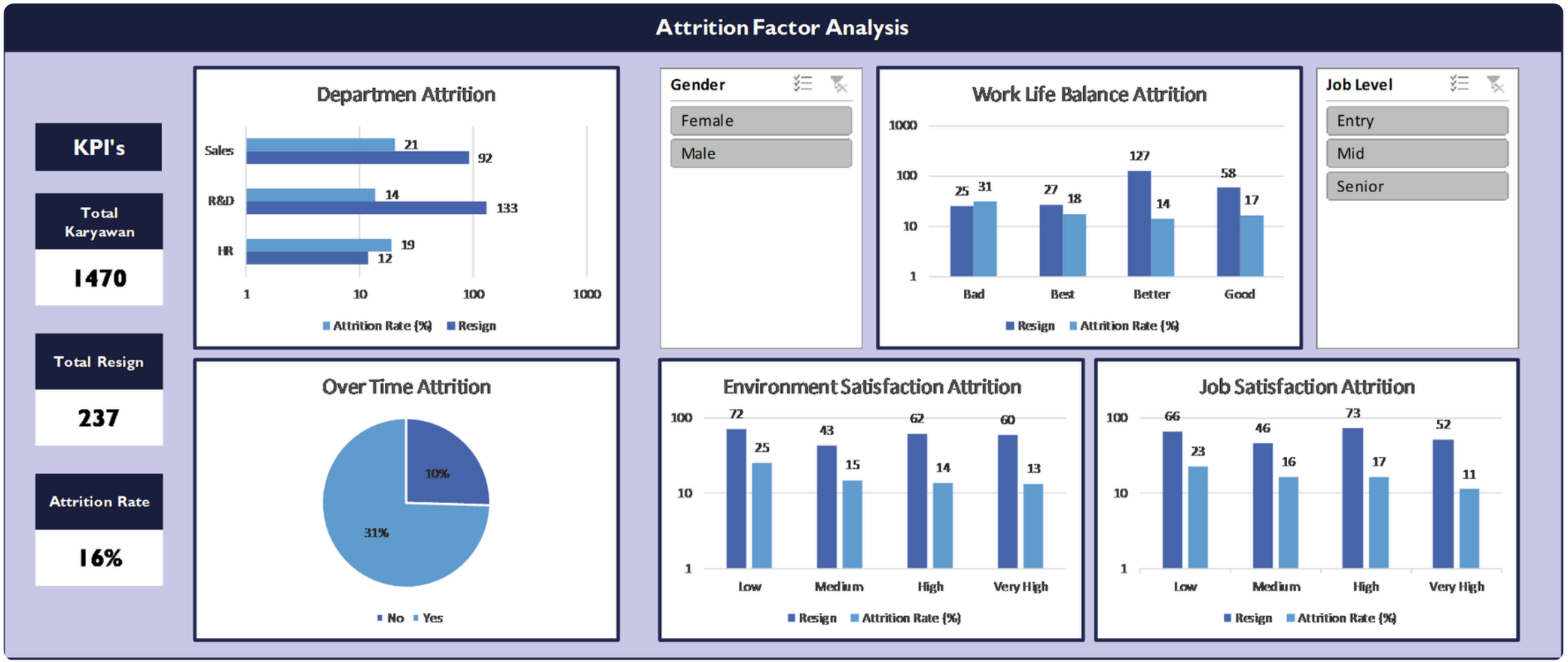
TOOLS 🛠️

- IF, VLOOKUP
- PivotTable, PivotChart
- Slicer (Gender, Job Role)

INSIGHT📌

- R&D mendominasi (65% dari total karyawan)
- Sales: performa tinggi (3.2) tapi kepuasan terendah (2.8)
- Dominan Laki-laki dengan presentase 60% laki-laki, 40% perempuan

Dashboard 2 - Attriction Factor Analysis



Dashboard 2 Insight

Key Performance Indicator 📊

- Total Resign: 237
- Attrition Rate: 16%

TOOLS 🛠️

- IF, VLOOKUP
- PivotTable, PivotChart, Calculated Field
- Slicer (Gender, Job Role)

INSIGHT📌

- OVERTIME jadi faktor TERTINGGI: 31% attrition (3× lipat!)
- SALES departemen paling berisiko 21% resign rate
- Job Satisfaction level 1: 23% vs level 4: 11%
- Environment Satisfaction level 1 25% (tertinggi)

Excel Mastery & Key Findings

Excel Mastery 📊

- PivotTables, PivotCharts, Slicers
- Calculated Field (AttritionRate)
- VLOOKUP (transformasi label), IF (kolom bantuan)

BUSINESS INSIGHTS 📈

- Demografi: R&D dominasi 65% karyawan, Sales punya performa tinggi tapi kepuasan terendah
- Attrition: Overtime faktor terkuat (31%, 3× lipat), Sales dept paling berisiko (21%)

KEY FINDINGS 💡

- Dashboard 1: "Sales: performa tinggi (3.2) tapi kepuasan terendah (2.8) - early warning attrition."
- Dashboard 2: "Overtime naikkan risiko resign 3× lipat. Evaluasi kebijakan lembur, terutama di Sales."

Thank you

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